JACOBO MATEO BEDOYA OQUENDO

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Backend developer since 2016 with a Mechanical Engineering background and a Master's in Automation & Industrial Control. Specialized in building high-performance real-time systems with Python and Go. Experience spans IoT data pipelines, distributed storage systems, real estate analysis engines, and industrial monitoring platforms.

PUBLIC PROJECTS:

JaDFS *(23/11/2025 - present)*
A distributed file storage system in Go, progressing from a simple file server to a fully distributed, fault-tolerant storage system.
view code

DependaMan *(09/03/2026 - 25/05/2026)*
Python dependency analysis and visualization tool. Analyzes your project's internal module structure and produces an interactive HTML graph, surfacing cycles, possibly death module, hotspots and coupling issues.
view code

HandyMan *(18/05/2026 - present)*
TTF font manipulation software written in Go.
view code

CORE TECHNOLOGIES:

Programming Languages

- Python (Advanced)
- Go (Intermediate)
- C (Basic)

Databases

- Relational:
  MySQL, PostgreSQL, SQLite
- Non Relational:
  MongoDB

Applied Architecture & Patterns

- Architectural:
  Event-Driven, Publisher-Subscriber, Client-Server, Master-Slave, Peer-to-Peer
- Design Patterns:
  Decorator, Plugin, Facade, Observer

Cloud & Infrastructure

- AWS (S3, Lambda, VPS, Elastic Beanstalk)
- gCloud (Bucket Storage, CloudRun, CloudApp, VPS, BigQuery)
- Azure (Blob Storage)
- Docker, Docker-compose, Docker-Swarm

WORK EXPERIENCE:

SHUSH
Senior Backend Developer (Contractor) *(January 2025 - Current)*

- Developed and maintained the platform's backend using FastAPI, adding new features and improving its stability.
- Developed and maintained the platform's micro-services using gRPC, incorporating new features and improving their stability using Python.
    - Designed and implemented a pipeline-based architecture for request execution, increasing the efficiency and stability of the system's configuration-driven architecture.
- Developed and maintained the dynamic usage-based billing system for the different clients that use the platform using Python.
- Designed, developed and maintained network observability and session-lookup tooling for the CGNAT infrastructure of different carriers in Go.
    - Designed and implemented a simple pipeline-based tool with a processing capacity of up to 150 thousand messages per second.
    - Designed and implemented a more complex micro-services- and event-based tool with a processing capacity of up to 1 million messages per second.

EarthFuneral
Senior Backend Developer *(March 2025 - December 2025)*

- Developed and maintained the platform's backend using Django, adding new features and improving overall stability.

- Developed and maintained a Django backend with a real-time pipeline using MQTT, WebSockets, Redis, and Celery to process high-volume sensor data.
- Implemented a PostgreSQL/TimescaleDB multi-database architecture and automated MQTT-based alert system with Celery task execution and WebSocket notifications.
- Improved performance through query optimization, indexing, and Redis caching.

Endava
Senior Developer (*August 2022 – December 2024)*

Math Engine (Real Estate Loan Analysis)

- Developed a loan sizing and investment analysis engine; implemented JSON-based automated test generation and binary search optimization for maximum loan and minimum NOI calculations.
- Contributed to the open-source Go version of the package.

Point Cloud (Desktop Annotation Tool)

- Led event-driven architecture for compute-heavy features using multiprocessing and memory optimizations.
- Automated blueprint generation via a local server reacting to point cloud annotations.
- Built a thread-safe facade over the BOX Cloud SDK for file management and uploads.

Sophos Solutions
Solutions Developer II (*April 2022 – July 2022)*

Continue the development and scaling of a mine monitoring platform to mitigate geological risk.

ActiveOne
Development Analyst *(April 2021 - March 2022)*

- Developed and maintained a web platform for uploading and classifying scanned proceedings using AWS AI services.
- Built a streaming tool to transfer files between AWS and Azure storage.
- Implemented a logging framework using a base meta-class and decorators for cross-application function tracking.

LUJACOL SAS
Owner *(March 2020 - October 2022)*

- Founded and operated a logistics company, building a web portal for client order management, shipping tracking, and merchandise documentation.
- Automated client communications with email reminders and notifications, keeping operations lean with minimal headcount.

Institución Universitaria ITM
Data Analyst *(October 2016 - January 2021)*

- Built a web portal to monitor scholarship recipients, generating early alerts for low academic performance and dropout risk to trigger welfare department intervention.
- Developed a metric to measure the university's academic impact on students across performance tiers.

EDUCATION

Master in Automation and Industrial Control (Coursework completed) - ITM
Mechanical Engineer - Universidad Pontificia Bolivariana